died out—Cuvier's research put an end to this belief. By comparing the fossilized remains of mammoths with those of modern elephants, he demonstrated for the first time that not only were the African and Asian elephants distinct species, but that the anatomy of a mammoth was completely different from either of theirs. From this, he concluded that the mammoth was in fact an extinct species, and went on to identify other extinct animals, including the giant ground sloth and the American mastodon. By establishing extinction as a verifiable fact, Cuvier launched the discipline of vertebrate paleontology.

Animal reconstruction

Renowned for his ability to reconstruct animals from their remains, Cuvier developed a theory he called the "correlation of parts," which stated that the structure of an organ in an animal's body was related to its function, as were all the other organs. For example, all hoofed mammals were herbivores and therefore must have teeth and a digestive system that is appropriate for eating plants. Using this principle, Cuvier was able to reconstruct complete skeletons of extinct organisms from just their isolated bones. He was also the first to identify the fossil of a flying reptile in 1800, which he named *pterodactyl*.

> **"The component parts** must be such as to render possible **the whole living being."**
>
> **Georges Cuvier**, 1817

To explain extinction, Cuvier proposed the theory of catastrophism—that a series of sudden, catastrophic events in the Earth's history, such as flooding, had wiped out certain species. He used his own observations of changes in rock strata to support this. Although his theory was later disproved, in exhuming fossils from rock strata, Cuvier revealed a significant fact: the deeper the rock stratum, the older the fossils it contained. Cuvier's work indicated that the deepest fossils were thousands of centuries old and increased the age of the Earth far beyond the previously accepted age of 6,000 years.

Anatomical classification

Cuvier publicly supported catastrophism over the new theory of evolution and used his seminal work

***Cuvier studied** the skeletal remains of elephants and woolly mammoths, revealing key differences in tooth structure that identified them as distinct species and showed that the mammoth was extinct.*

PRIOR TO **CUVIER'S STUDIES** OF **FOSSILS,** SCIENTISTS AGED THE EARTH AT **6,000 YEARS**

HIS NAME IS ONE OF **72** INSCRIBED ON THE **EIFFEL TOWER**